Aakanksha Trivedi

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EDUCATION

Master of Science in Computer Science (GPA: 4.00/4.00) Jan 2025 - Dec 2026
Purdue University Northwest Hammond, USA

Bachelor of Engineering - Computer Engineering (CGPA: 9.18/10) Oct 2020 - Jun 2024
Institute of Engineering & Technology, Devi Ahilya University Indore, India

SKILLS

Languages: C/C++, Python, Java, C#, JavaScript, PHP, SQL, HTML
Systems & Networking: Linux, Kernel Programming, eBPF, XDP, libbpf, TCP/IP, Modbus/TCP, Network Namespaces
Technologies: Machine Learning, RAG, LangChain, FAISS, Ollama, Streamlit, Redis, GraphQL, Shopify API, n8n
Tools: Docker, MySQL, Wireshark, tcpdump, bpftool, Clang/LLVM, GitHub, Visual Studio Code, Google Colab

PROFESSIONAL EXPERIENCE

Software Developer Intern, *Beulah Inc* May 2026 - Aug 2026
- Developed PHP/MySQL application features and Dockerized n8n workflows to automate catalog, file, and part-management operations. Valparaiso, USA
- Integrated Shopify GraphQL APIs for product/variant updates, inventory mapping, metafields, staged image uploads, collections, and Shopify ID writeback.
- Implemented sync-state tracking, failure branches, error logging, rerun-safe processing, SQL validation, and production debugging across legacy modules.

Student Researcher, *Purdue University Northwest* Aug 2026 - Present
- Developing an eBPF/XDP intrusion-detection pipeline for Modbus/TCP industrial-control traffic with protocol-aware parsing directly in the Linux networking path. Hammond, USA
- Extracting transaction IDs, function codes, register references, packet lengths, malformed/stacked frames, duplicate transactions, and request-rate behavior via XDP.
- Built a libbpf ring-buffer collector and Linux namespace/veth testbed for CIC Modbus attack analysis, time-window feature generation, throughput, and latency benchmarking.

Student Researcher, *Purdue University Northwest (CIVS)* Jan 2026 - May 2026
- Built a GPU-accelerated Unity digital twin to visualize real-time thermal behavior in continuous steel casting from dynamic CSV-driven simulation data. Hammond, USA
- Developed automated thermal segmentation and annotation using configurable temperature thresholds to identify critical casting regions and improve interpretability.
- Implemented synchronized 2D/3D transverse, longitudinal, curved, and volumetric rendering pipelines with real-time data synchronization and interactive exploration.

Research and Teaching Assistant, *Purdue University Northwest* Jun 2025 - May 2026
- Conducted literature reviews and comparative research across machine learning, cybersecurity, and quantum computing while standardizing reproducible research workflows. Hammond, USA
- Co-authored and published IEEE MASS 2025 research on quantum-enhanced clustering for intrusion detection in IoT environments.
- Mentored undergraduate Java labs by debugging code, teaching object-oriented programming concepts, evaluating assignments, and providing code-review support.

Research Intern, *Indian Institute of Technology (IIT) Indore* May 2024 - Nov 2024
- Developed a low-latency Linux eBPF/XDP intrusion-detection system for real-time packet inspection and protocol-aware traffic analysis in the kernel. Indore, India
- Architected a scalable deep-packet inspection pipeline chaining 140+ eBPF programs with tail calls to stay within verifier instruction limits.
- Implemented HTTP/FTP/SSH payload-signature matching and Slowloris detection, then benchmarked XDP execution modes while minimizing kernel-user context switching.

PROJECTS

Lifeline AI, *Disaster Management Dashboard for First Responders* Jun 2026 - Present
- Built an AI-powered disaster-management dashboard to help first responders monitor incidents, prioritize events, and coordinate emergency-response information.
- Integrated Browserbase API, Deepgram API, and Redis to support web automation, speech-to-text updates, and low-latency real-time information retrieval.
- Collaborated with a hackathon team to design, implement, and present a working prototype focused on improving first-responder situational awareness.

Multi-Agent Research Assistant (RAG + LLM System), Mar 2026 - May 2026
- Architected a multi-agent RAG system orchestrating specialized agents for document retrieval, web search, and LLM-based response generation across multiple sources.
- Implemented embedding-based semantic retrieval using FAISS, LangChain, and Ollama with vector similarity search and citation-aware response generation.
- Built a real-time Streamlit interface for document ingestion, vector indexing, and interactive querying across multiple indexed documents.

HACKATHONS

UC Berkeley AI Hackathon 2026 UC Berkeley, USA
Cal Hacks 12.0 San Francisco, USA
Smart India Hackathon 2022 and 2023 Finalist India

PUBLICATIONS AND AWARDS

Quantum-Enhanced Clustering for Intrusion Detection in IoT Devices, *IEEE MASS 2025* Aug 15, 2025
Graduate Research Award, *Purdue University Northwest* Aug 25, 2025